

TellMeLight Rev A3 JLC-oriented schematic baseline

USB-C J1: VBUS enters protected 5V rail; CC1/CC2 use 5.1k Rd pull-downs.

USB data: J1 D+/D- -> TPD2EUSB30 U5 -> 27R series resistors -> RP2040 U1 USB pins.

Power: VBUS -> AP2112K-3.3 U4 -> 3V3 for RP2040, flash, and LP5024 logic.

MCU: RP2040 U1 uses W25Q32JVSSIQ U3 QSPI flash, 12 MHz Y1, pogo SWD pads, BOOT SW1, RESET SW2.

LED driver: RP2040 I2C0 SDA/SCL -> LP5024 U2 with 4.7k pull-ups to 3V3.

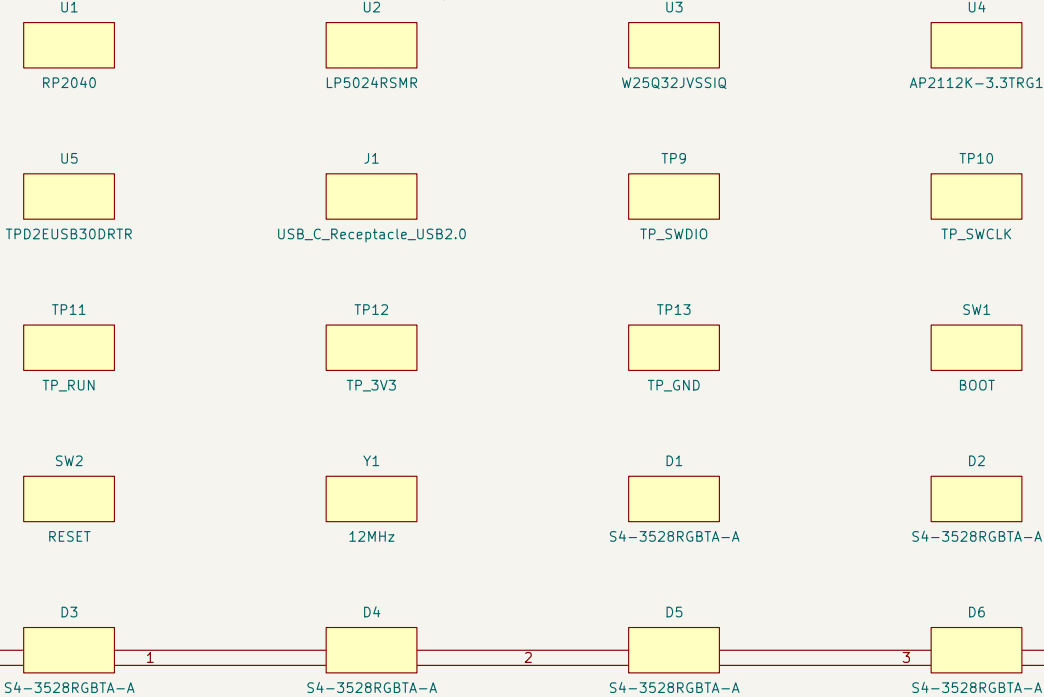
Light zones: LP5024 OUT0..OUT17 sink six common-anode 3528 RGB LEDs D1..D6 from VLED.

Reserved: LP5024 OUT18..OUT23 are kept for Rev B brightness or status expansion.

Manufacturing: 4-layer board, double-sided SMT, no assumed user hand-soldering.

Rev A3 pin map is documented in hardware/notes/rev-a2-pin-map.md; RGB LED pinout remains a RED order blocker.

Review symbols below represent selected Rev A3 components; wiring still follows the net-plan notes above.



JLC-oriented block-level schematic generated by tools/hardware/generate-rev-a1-kicad.mjs		
USB-C RP2040 + LP5024 six-zone AI session light		
TellMeLight		
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